

jkl - Answer Key

1. What are the two broad categories of cells?

Answer: C

Explanation: The textbook states that there are two broad categories of cells: prokaryotic and eukaryotic.

2. What is the primary function of the cell membrane?

Answer: To control what enters and exits the cell

Explanation: The cell membrane is a selectively permeable barrier that separates the cell's interior from its external environment and controls the movement of substances in and out of the cell.

3. Cellular respiration occurs mainly in the nucleus.

Answer: False

Explanation: Cellular respiration occurs mainly within the mitochondria, often called the powerhouse of the cell.

4. What is the fluid mosaic model?

Answer: A structure that describes the cell membrane as a phospholipid bilayer embedded with proteins

Explanation: The fluid mosaic model is a description of the cell membrane's structure, which is composed primarily of a phospholipid bilayer embedded with proteins.

5. What is the process by which cells convert nutrients into usable energy?

Answer: B

Explanation: Cellular respiration is the process by which cells convert nutrients, primarily glucose, into usable energy in the form of ATP.

6. What are the three main stages of cellular respiration?

Answer: Glycolysis, the Krebs cycle, and the electron transport chain

Explanation: The textbook states that cellular respiration consists of three main stages: glycolysis, the Krebs cycle, and the electron transport chain.

7. Glycolysis is an aerobic process.

Answer: False

Explanation: Glycolysis is an anaerobic process, meaning it does not require oxygen.

8. What is chemiosmosis?

Answer: A process that drives the production of a large majority of the cell's ATP through the electron transport chain

Explanation: Chemiosmosis is the process by which the electron transport chain produces a large majority of the cell's ATP, and it requires oxygen as the final electron acceptor.